

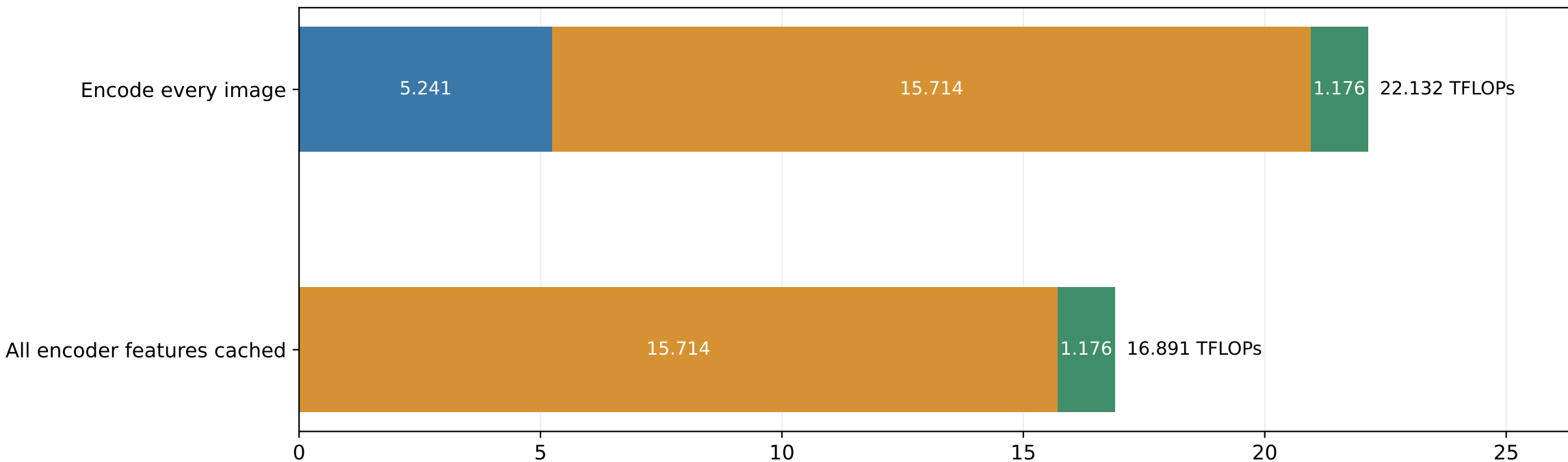
Qwen3-VL-4B: image encoding precedes language prefill



4 images, each 640 x 640
400 merged positions per image

2,000 prompt positions
Last logits produce output token 1

127 single-token calls
KV grows to 2,127 positions



Matrix work per request (10¹² FLOPs); not TFLOP/s or latency

Same request: four 640 x 640 images + 400 other prompt positions, 128 output tokens.

Encoder-cache hits remove only encoding work. Feature reads, transfers and CPU preprocessing are outside these bars.